

Client Requirements

1. Project Information

ITEM	DETAILS
Module	CMPG 325 – Computer Networks
Project ID	CMPG325-2026-134
Client ID	CLI-134
Client	Barolong Auto Dealership
Location	Mahikeng
Industry	Automotive
Allocated Address Block	172.30.88.0/23
Networking Challenge	STP – Loop Prevention & Root Design

2. Client Overview

Barolong Auto Dealership is an automotive dealership located in Mahikeng. The purpose of this project is to design a computer network that provides reliable connectivity to the different areas of the dealership.

The proposed network will provide connectivity for dealership users and devices while separating different types of network traffic through logical network segmentation.

The network will be designed and simulated using Cisco Packet Tracer.

3. Network Requirements

The proposed network must satisfy the following requirements:

1. Provide reliable network connectivity throughout the dealership.
2. Provide appropriate connectivity for the different dealership departments.
3. Use the allocated IPv4 addressing block **172.30.88.0/23**.
4. Provide an appropriate combination of routers, switches and end devices.
5. Provide logical separation of network traffic using VLANs.
6. Provide wireless connectivity in the specified public areas.
7. Prevent Layer 2 switching loops through the use of Spanning Tree Protocol (STP).
8. Provide an appropriate STP root bridge design.
9. Provide redundant network paths where appropriate.
10. Provide an addressing/design accommodation for the planned branch office.
11. The planned branch office must not be implemented as a second physical site.
12. The completed design must be suitable for implementation and testing in Cisco Packet Tracer.

4. Wireless Requirement

Wireless coverage is required only in the specified public areas.

The wireless network will therefore be designed primarily for public/guest access rather than providing wireless coverage throughout the entire dealership.

The public wireless network will be logically separated from the internal dealership networks.

5. Networking Challenge

The assigned networking challenge is:

STP – Loop Prevention & Root Design

The network will contain redundant Layer 2 connections between switches. These redundant connections improve network resilience but can create Layer 2 switching loops.

Spanning Tree Protocol will therefore be used to prevent switching loops and to control which switch becomes the root bridge.

The proposed design will use a primary and secondary root switch so that STP operation is predictable and can be demonstrated during testing.

6. Client Change Request – CR6

The client has a planned future branch office.

The project brief specifies that this change request requires:

Design and addressing accommodation only. No second-site build is required.

The proposed network will therefore reserve an IP address range for the future branch office without creating a second physical branch-office topology in Packet Tracer.

7. Design Assumptions

The following assumptions are made for the initial design:

- The dealership requires separate logical networks for administration, sales, workshop/service and public wireless access.
- Network management traffic will be separated from normal user traffic.
- The public wireless network will be isolated from internal dealership networks.
- Redundant switch connections will be included to demonstrate the assigned STP challenge.
- The future branch office will be accommodated through reserved addressing only.
- Exact numbers of users and devices may be adjusted during detailed implementation if additional requirements are provided.

8. Expected Outcome

The final network should provide a structured, scalable and reliable network environment for Barolong Auto Dealership.

The design should demonstrate appropriate IP addressing, network segmentation, redundant switching paths and STP loop prevention while complying with the wireless and branch-office constraints provided by the client.